

Customer Shopping Behavior Analysis

Uncovering spending patterns, customer segments, product preferences, and subscription behavior from **3,900 transactions** to guide strategic business decisions.



Dataset at a Glance

3,900

Transactions

18

Features

37

Missing Ratings

Key Feature Groups

- **Demographics:** Age, Gender, Location, Subscription Status
- **Purchase Details:** Item, Category, Amount, Season, Size, Color
- **Behavior:** Discounts, Promo Codes, Frequency, Review Rating, Shipping

Data Preparation in Python

01

Load & Explore

Imported dataset with pandas; used `df.info()` and `.describe()` for initial profiling.

02

Clean & Engineer

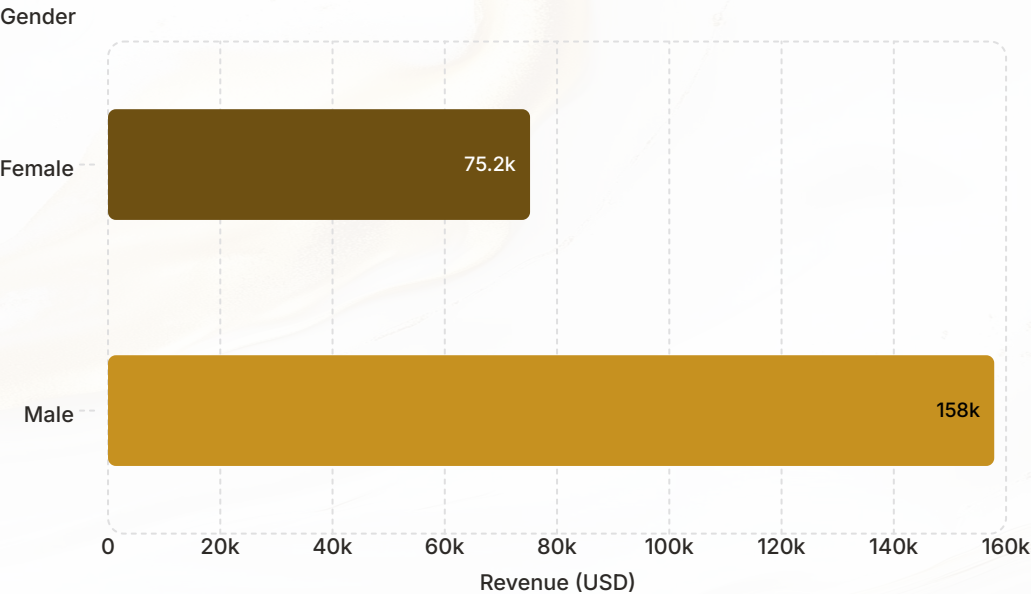
Imputed missing Review Ratings with category medians; created `age_group` and `purchase_frequency_days` columns.

03

Standardize & Load

Renamed columns to snake_case; dropped redundant `promo_code_used`; loaded cleaned data into PostgreSQL.

Revenue & Spending Insights

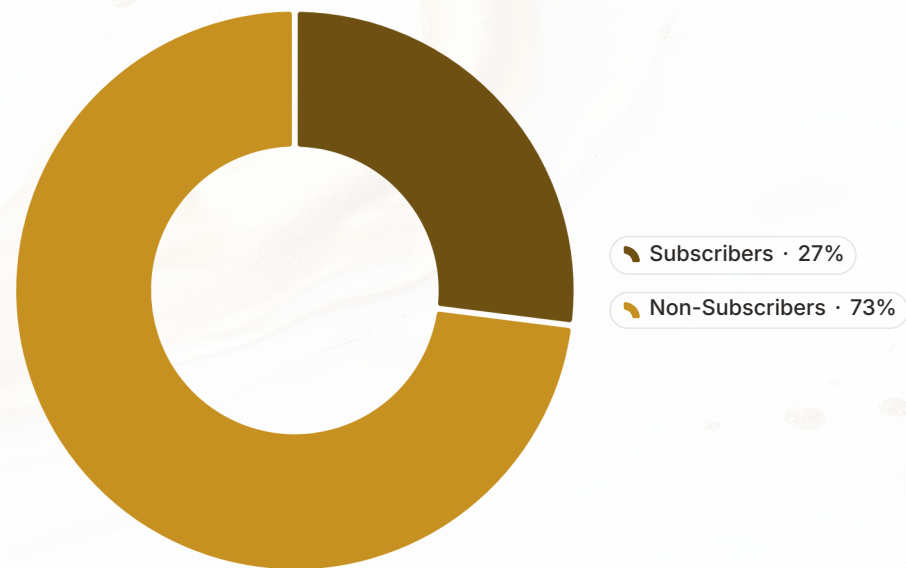


Shipping Type vs. Spend

Shipping Type	Avg. Purchase (USD)
Standard	\$58.46
Express	\$60.48

Express shipping users spend slightly more on average, suggesting a higher-value customer segment worth targeting.

Subscriber vs. Non-Subscriber Breakdown



Revenue Comparison

Status	Avg Spend	Total Revenue
Subscribers	\$59.49	\$62,645
Non-Subscribers	\$59.87	\$170,436

Non-subscribers represent **73%** of customers — a major opportunity to convert through targeted subscription campaigns.

Top-Rated Products & Discount Dependency

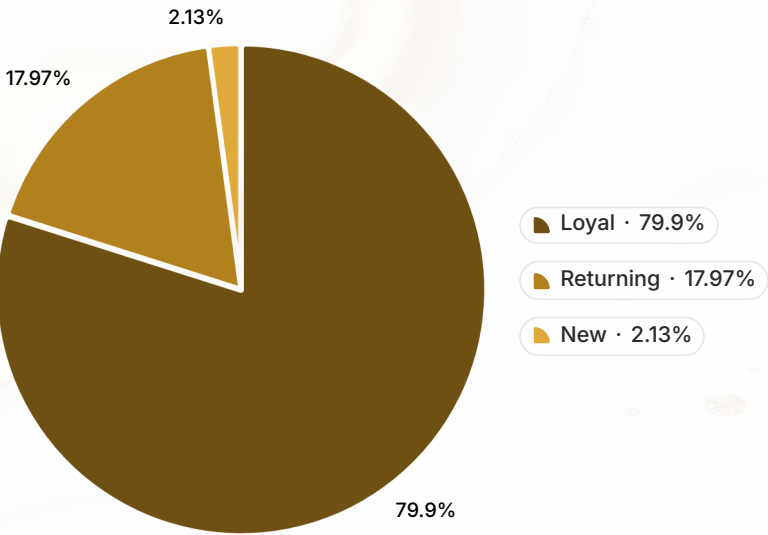
Top 5 Products by Avg. Rating

Product	Avg. Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discount-Dependent Products

Product	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Customer Segmentation & Loyalty



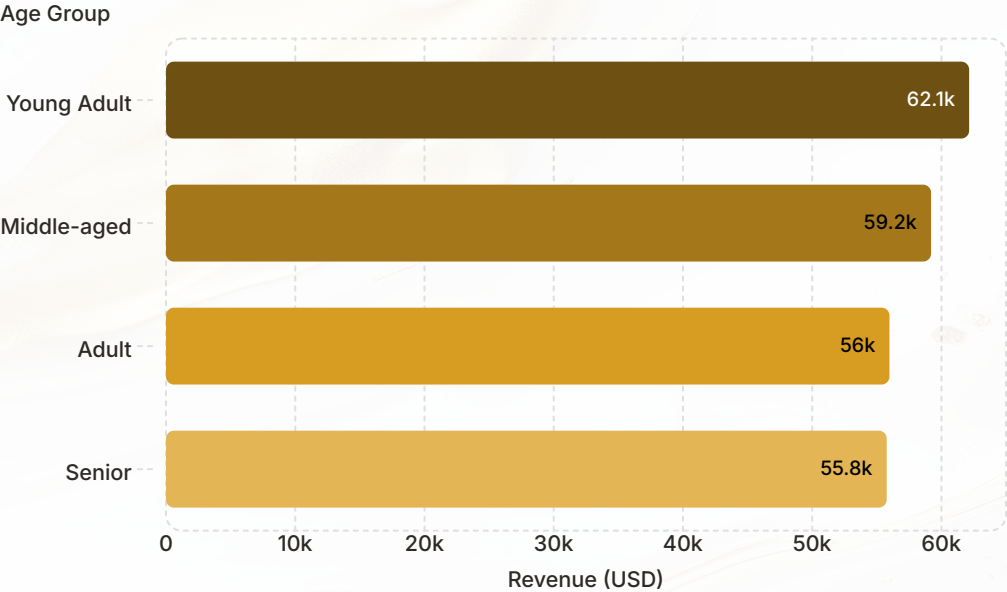
Repeat Buyers & Subscriptions

Among customers with **more than 5 purchases**:

Subscription	Repeat Buyers
No	2,518
Yes	958

The vast majority of loyal customers are **not subscribed** — a key conversion opportunity.

Revenue by Age Group & Top Products by Category



Top Products per Category

Category	#1 Product	Orders
Accessories	Jewelry	171
Clothing	Blouse / Pants	171
Footwear	Sandals	160
Outerwear	Jacket	163

Power BI Dashboard

3.9K

Customers

\$59.76

Avg. Purchase

3.75

Avg. Rating

Interactive filters by Subscription Status, Gender, Category, and Shipping Type. Visualizations include revenue by category, sales by age group, and subscription distribution.

Business Recommendations



Boost Subscriptions

Promote exclusive benefits to convert the 73% non-subscriber base.



Loyalty Programs

Reward repeat buyers to accelerate movement into the Loyal segment.



Review Discount Policy

Balance sales boosts with margin control on high-discount products.



Targeted Marketing

Focus on Young Adults and express-shipping users — highest revenue contributors.